



Name:

Teacher:

SCEGGS Darlinghurst

HSC Assessment 2
Tuesday, 6th June, 2006

Extension 1 Mathematics

General Instructions

- Time allowed – 75 minutes
- Weighting 35%
- This paper has **four** questions
- Attempt **all** questions and show all necessary working
- Marks may be deducted for careless or badly arranged work
- Write using blue or black pen, diagrams in pencil
- Write your name and your teacher's name at the top of each page
- Approved calculators, mathematical templates and geometrical instruments may be used
- A table of standard integrals is provided at the back of this paper

Questions	Total	Comm.	Reas.	Calc.
1	/12	/2	/1	/5
2	/10		/2	/8
3	/12	/2	/3	/1
4	/13	/1	/6	/4
TOTAL	/47	/5	/12	/18

Question 1 (12 marks)	Marks
(a) Find the inverse function of $y = 3 + \log_e x$	1
(b) Differentiate $\cos^{-1}(2x)$	1
(c) Evaluate $\int_0^{2\sqrt{3}} \frac{dx}{4+x^2}$	2
(d) Find $\int \frac{dx}{\sqrt{9-4x^2}}$	2
(e) Consider the function $f(x) = x^2 - 2x$	
i. Sketch $y = f(x)$ clearly indicating the coordinates of the vertex and any intercepts. Use the same scale on both axes.	1
ii. State the largest domain that includes the origin for which the function has an inverse function.	1
iii. State the domain and range of this inverse function, $f^{-1}(x)$.	2
iv. Sketch $y = f^{-1}(x)$ on the same set of axes as in part(i). Label the two graphs clearly.	1
v. Find the gradient of $y = f^{-1}(x)$ at the origin.	1

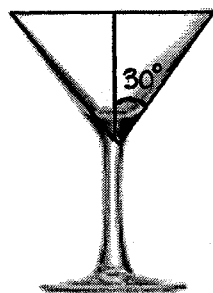
Question 2 begins on page 2 ...

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Question 2 (10 marks)

Marks

- (a) Using the substitution $u = x + 1$, evaluate $\int_1^3 \frac{x-1}{(x+1)^3} dx$ 4
- (b) A cocktail glass is in the shape of an inverted right-cone with semi-vertical angle 30° . It is initially filled with liquid to a height h cm.



- i. Show that the volume of liquid in the glass is given by: 2

$$V = \frac{1}{9}\pi h^3$$

- ii. The liquid is drunk (through a straw) at a rate of 1 mL/s. Find 4
the rate at which the height of the liquid is changing when there is
only 10 mL of liquid left. Answer to 2 decimal places.
[Note: $1 \text{ cm}^3 = 1 \text{ mL}$]

Question 3 begins on page 3 ...

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Question 3 (12 marks)

Marks

- (a) Find the exact value of

3

$$\cos \left(\sin^{-1} \left(\frac{1}{3} \right) + \cos^{-1} \left(-\frac{2}{5} \right) \right)$$

- (b) A hot cup of coffee loses heat in a colder environment according to Newton's Law of Cooling, $\frac{dT}{dt} = -k(T - T_e)$, where t is time in minutes, k is a constant, and T_e is the temperature of the environment.

- i. Show that $T = T_e + Ae^{-kt}$ is a solution of this equation, for some constant A . 1

At 7 am on a cold morning (10°C) I buy a cup of coffee. The coffee cools from 75°C to 65°C in 15 minutes.

- ii. Find the exact value of the constants A and k . [Note that $T_e = 10$] 2
- iii. What is the temperature of the coffee when I arrive at work at 7:20 am? (Answer to the nearest degree) 1
- iv. How long do I have to drink my coffee before the temperature falls below 55°C and is too cold to drink? (Answer to the nearest minute) 1
- v. At what rate was the tea cooling initially? 2
- vi. Sketch a graph of the temperature of the coffee against time. Label any important features. 2

Question 4 begins on page 4 ...

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Question 4 (13 marks)

Marks

(a) Consider the function $f(x) = 3 \cos^{-1} \frac{x}{2}$

- i. State the domain and range of $y = f(x)$ 2
- ii. Neatly sketch $y = f(x)$ 1
- iii. Find the area of the region in the first quadrant bounded by the curve $y = f(x)$ and the co-ordinate axes. 3

- (b) i. Using the substitution $u = 1 - x^2$, or otherwise, find 3

$$\int \frac{x}{\sqrt{1-x^2}} dx$$

- ii. Differentiate $x \sin^{-1} x$ 1
- iii. Hence find 3

$$\int_0^{\frac{1}{2}} \sin^{-1} x dx$$

END OF ASSESSMENT

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Question 1
 Know /1
 Calc. /5
 Comm. /2 (12 marks)

(a) Original: $y = 3 + \log_e x$

Inverse: $x = 3 + \log_e y$

$x - 3 = \log_e y$

$y = e^{x-3}$ ✓

Once swap x & y ,
 Must make y subject

(b) $y = \cos^{-1} 2x$

$y' = \frac{-2}{\sqrt{1-4x^2}}$ ✓ calc.

Lots of people forgot
 - chain rule
 - negative

(c) $\int_0^{2\sqrt{3}} \frac{dx}{4+x^2}$

$= \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^{2\sqrt{3}}$ ✓

- Lots of people forgot
 the $\frac{1}{2}$ out front

$= \frac{1}{2} \tan^{-1} \sqrt{3} - \frac{1}{2} \tan^{-1} 0$

$= \frac{\pi}{6}$ ✓ calc.

- Must evaluate
 $\tan^{-1} \sqrt{3}$ exactly.
 Did not accept a
 rounded decimal.

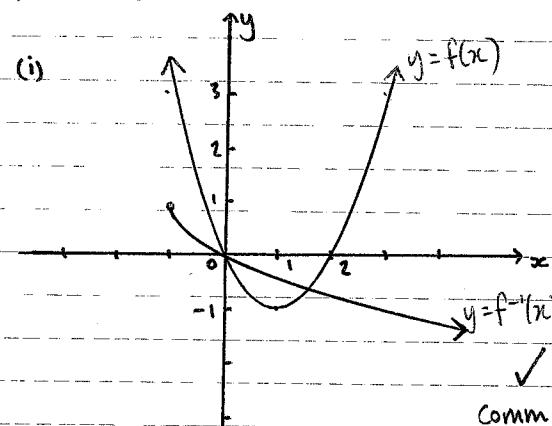
(d) $\int \frac{dx}{\sqrt{9-4x^2}}$

$= \int \frac{dx}{\sqrt{3^2 - (2x)^2}}$

$= \frac{1}{2} \sin^{-1} \left(\frac{2x}{3} \right) + C$ ✓ ✓ calc.

- People forgot the $\frac{1}{2}$
 (Reverse Chain Rule)

(e) $f(x) = x^2 - 2x$



- The inverse needs
 to go through (3, -1)
 if it was well off
 you didn't get the
 mark

- Shouldn't need a
 page of calculus to
 draw a parabola

(ii) $x \leq 1$ ✓

Generally well done

(iii) $f^{-1}(x)$: D: $x \geq -1$, x real ✓
 R: $y \leq 1$, y real ✓

(iv) (shown above) ✓ comm.

(v) original: $y = x^2 - 2x$

inverse: $x = y^2 - 2y$

$\frac{dx}{dy} = 2y - 2$

$\frac{dy}{dx} = \frac{1}{2y-2}$

@ (0,0)

$m_T = \frac{1}{2 \times 0 - 2}$

$= -\frac{1}{2}$ ✓ Reas

People either knew it
 or they didn't.

Question 2

Reas. /2
Calc. /8 (10 marks)

$$(a) \int_1^3 \frac{x-1}{(x+1)^3} dx$$

$$u = x+1$$

$$du = dx$$

$$= \int_2^4 \frac{u-2}{u^3} du \checkmark$$

$$x=1 \rightarrow u=2$$

$$x=3 \rightarrow u=4 \checkmark$$

$$= \int_2^4 u^{-2} - 2u^{-3} du$$

$$= \left[\frac{u^{-1}}{-1} - \frac{2u^{-2}}{-2} \right]_2^4 \checkmark$$

$$= \left[-\frac{1}{u} + \frac{1}{u^2} \right]_2^4$$

$$= \left(-\frac{1}{4} + \frac{1}{16} \right) - \left(-\frac{1}{2} + \frac{1}{4} \right)$$

$$= \frac{1}{16} \checkmark$$

(Calc4.)

This part was well done.

You will not be given
any marks for integrating
incorrectly and substituting
into nonsense.



$$V_{\text{cone}} = \frac{1}{3} \pi r^2 h$$

$$\tan 30^\circ = \frac{r}{h}$$

$$r = h \tan 30^\circ$$

$$r = \frac{h}{\sqrt{3}} \checkmark$$

$$\Rightarrow V = \frac{1}{3} \pi \left(\frac{h}{\sqrt{3}} \right)^2 h$$

$$\Rightarrow V = \frac{\pi h^3}{9} \checkmark$$

(Reas.2)

$$(ii) \frac{dV}{dt} = 1 \frac{\text{mL}}{\text{s}}, \frac{dh}{dt} = ? \text{ when } V=10$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

↓
find
this!

Clearly show this
expression for the radius.

AND substitute it into
the volume of a cone
formula.

$$V = \frac{1}{3} \pi h^3$$

$$\frac{dV}{dh} = \frac{1}{3} \pi h^2$$

$$\frac{dh}{dV} = \frac{3}{\pi h^2} \checkmark$$

$$\therefore \frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$= \frac{3}{\pi h^2} \times 1$$

$$= \frac{3}{\pi h^2} \checkmark$$

$$\text{When } V=10$$

$$\frac{1}{3} \pi h^3 = 10$$

$$h = \sqrt[3]{\frac{30}{\pi}} \checkmark$$

$$\therefore \frac{dh}{dt} = \frac{3}{\pi \left(\sqrt[3]{\frac{30}{\pi}} \right)^2} = 0.10199...$$

$$= 0.10 \text{ to } 2 \text{ d.p.}$$

$\therefore$ The height of the liquid is
changing at 0.10 cm/s $\checkmark$

(Calc4)

This part was well done.

Many students forgot
that 10 was volume and
not the height.

Read the units very
carefully.

$$10 \text{ mL} = 10 \text{ cm}^3$$

$\therefore$ It is a volume
not a height.

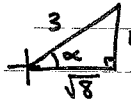
Question 3

Calc 11
Comm 12
Reas 13 (12 marks)

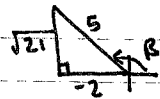
(a) $\cos(\sin^{-1} \frac{1}{3} + \cos^{-1} \frac{-2}{5})$

$\downarrow$
 α $\downarrow$
 β

let $\alpha = \sin^{-1} \frac{1}{3}$



let $\beta = \cos^{-1} \frac{-2}{5}$



$\therefore \cos(\sin^{-1} \frac{1}{3} + \cos^{-1} \frac{-2}{5})$

$= \cos(\alpha + \beta)$

$= \cos \alpha \cos \beta - \sin \alpha \sin \beta$ ✓

$= \frac{\sqrt{8}}{3} \times \frac{-2}{5} - \frac{1}{3} \times \frac{\sqrt{21}}{5}$ ✓

$= \frac{-2\sqrt{8} - \sqrt{21}}{15}$ Reas.

(b)(i) $T = T_e + Ae^{-kt}$

$\frac{dT}{dt} = A \times e^{-kt} \times -k$

$= -k \times (Ae^{-kt})$

$= -k(T - T_e)$ ✓ Calc

(ii) $t=0$ $T=75^\circ\text{C}$

$\Rightarrow 75 = 10 + Ae^{-k \times 0}$

$A = 65$ ✓

$t=15$ $T=65^\circ\text{C}$

$\Rightarrow 65 = 10 + 65e^{-k \times 15}$

$55 = 65e^{-15k}$

$k = \frac{-1}{15} \log_e \left(\frac{55}{65} \right)$ ✓

$(k \doteq 0.0111369...)$

ok, except for some careless errors

two people had $\sqrt{21}$ turn into $\sqrt{2}$ for example!

These questions are easy to check by using your calculator! No one should be losing marks for carelessness in these type of questions.

Well done.

(iii) $T = 10 + 65e^{-kt}$

$t=20$ $T=?$

$T = 10 + 65 \times e^{-k \times 20}$

$= 62.021...$

$\doteq 62^\circ$ (to nearest degree) ✓

(iv) $T=55$ $t=?$

$55 = 10 + 65e^{-kt}$

$45 = 65e^{-kt}$

$t = \frac{-1}{k} \log_e \left(\frac{45}{65} \right)$

$= 33.018...$

$\doteq 33$ minutes ✓

Falling below 55°C does not mean solve for 54° (54.9°C is still below!).

(v) $t=0$ $\frac{dT}{dt} = ?$

$\downarrow$
 $T=75$

$\frac{dT}{dt} = -k(T - T_e)$

$= -k(75 - 10)$ ✓

$= -65k$

$= -0.7239...$

$\doteq -0.724^\circ\text{C/min}$ ✓

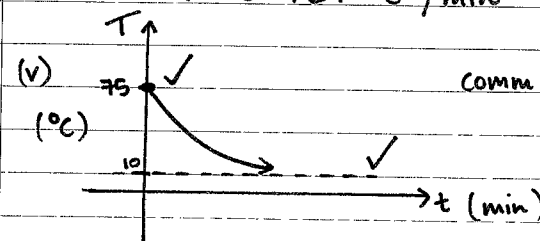
Make sure you include units!!!

(& apologies for the typo)

or, $\frac{dT}{dt} = -65ke^{-kt}$

$= -65k$ when $t=0$

$\doteq -0.724^\circ\text{C/min}$



Lumpy graphs did not get full marks. Plotting points if fine (& good) but make sure they can be joined with a smooth curve & the scale allows it to approach the asymptote.

Question 4

Comm. /1
Reas. /6
Calc. /4 (13 marks)

(a) $f(x) = 3 \cos^{-1}\left(\frac{x}{2}\right)$

(i) $\left(\frac{y}{3}\right) = \cos^{-1}\left(\frac{x}{2}\right)$

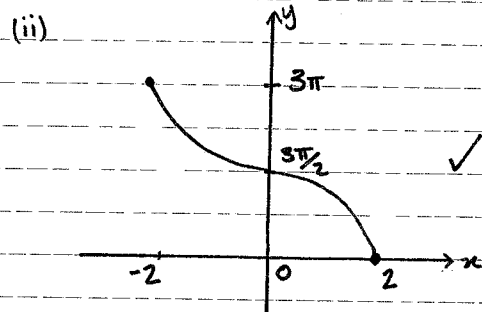
R: $0 \leq \frac{y}{3} \leq \pi$

R: $0 \leq y \leq 3\pi$ ✓

D: $-1 \leq \frac{x}{2} \leq 1$

D: $-2 \leq x \leq 2$ ✓

Well done!



✓ Comm.

The value of the y-intercept at $y = 3\pi/2$ is very important.

(iii) $y = 3 \cos^{-1}\left(\frac{x}{2}\right)$

$\left(\frac{y}{3}\right) = \cos^{-1}\left(\frac{x}{2}\right)$

$x = 2 \cos\left(\frac{y}{3}\right)$

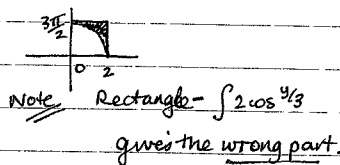
Area = $\int_0^{3\pi/2} 2 \cos \frac{y}{3} dy$ ✓

$= \left[\frac{2 \sin\left(\frac{y}{3}\right)}{\frac{1}{3}} \right]_0^{3\pi/2}$ ✓

$= \left[6 \sin\left(\frac{y}{3}\right) \right]_0^{3\pi/2}$

$= 6 \sin\left(\frac{\pi}{2}\right) - 6 \sin(0)$

$= 6$ ✓ Reas.



Make sure you divide the $\frac{1}{3}$
not multiply by it.

(b) (i) $\int \frac{x}{\sqrt{1-x^2}} dx$ $u = 1-x^2$
 $du = -2x dx$
 $= -\frac{1}{2} \int \frac{du}{\sqrt{u}}$ ✓ $-\frac{1}{2} du = x dx$
 $= -\frac{1}{2} \int u^{-1/2} du$
 $= -\frac{1}{2} \times \frac{u^{1/2}}{1/2} + C$ ✓
 $= -\sqrt{u} + C$
 $= -\sqrt{1-x^2} + C$ ✓ Calc.

do not mix up the letters
in your substitution lines.

The final answer must
be written in terms of x .

Note $\sqrt{1-x^2} \neq 1-x$
Never, ever, ever!

(ii) $\frac{d}{dx} (x \sin^{-1} x)$ Use the product rule

$= x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x$

$= \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x$ ✓ Calc.

product rule
 $uv' + vu'$

(iii) $\int_0^{1/2} \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x dx = \left[x \sin^{-1} x \right]_0^{1/2}$
 $+ \int_0^{1/2} \sin^{-1} x dx = \left[x \sin^{-1} x \right]_0^{1/2} - \int_0^{1/2} \frac{x}{\sqrt{1-x^2}} dx$

This is Reverse Product Rule.

The question states
hence

So you must use part (i) and (ii)

$= \left[x \sin^{-1} x \right]_0^{1/2} + \left[\sqrt{1-x^2} \right]_0^{1/2}$ ✓

$= \left[\frac{1}{2} \sin^{-1} \frac{1}{2} - 0 \right] + \left[\sqrt{1-\frac{1}{4}} - \sqrt{1-0} \right]$

$= \frac{1}{2} \times \frac{\pi}{6} + \sqrt{3/4} - 1$

$= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1$ ✓

Reas.

if the question stated
or otherwise then you
could use a different
method.